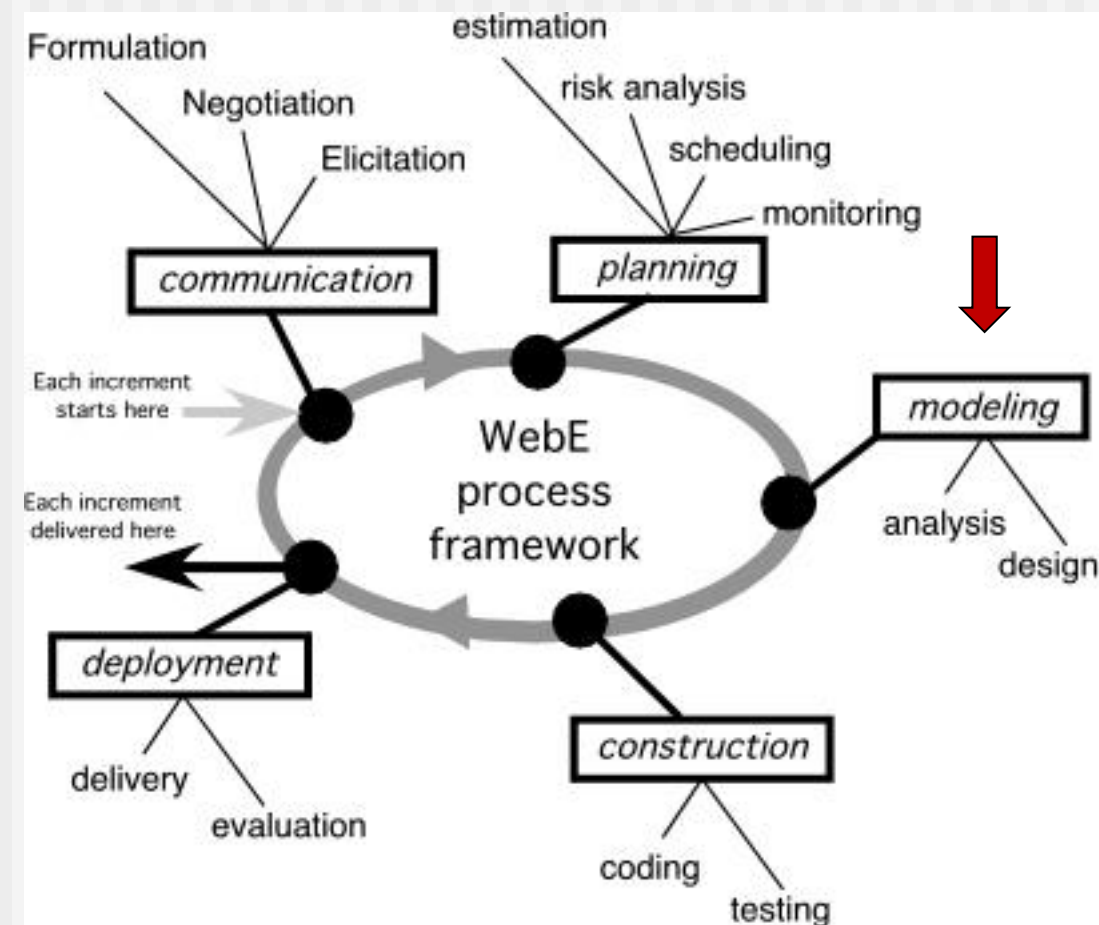

Web Development and Design

Lecture 05: Analysis Modeling for Web Applications

WebE Process Activities & Actions



The Modeling Activity

- One of the key WebE activities is modelling, and this includes two main actions: **analysis and design**.
 - **Analysis modeling** helps you to understand the nature of the problem being addressed and the “shape” of the WebApp that will allow you to address that problem .
 - **Design modeling** is about understanding the internal structure of the WebApp being developed and how this creates the shape of the WebApp that was identified by the analysis model.

The Modeling Activity

- There will be different points of view in the modelling for example:
 - **System Architect** will focus on the overall **structure** of the WebApp, the broad **technologies** that are being used, and how the WebApp **interfaces** to other systems.
 - **Information Architect** will emphasize the nature, structure, and **management of the content** within the WebApp (how the information on the website is organized).
 - **A programmer** on the WebE team will be most interested in the lower-level functional detail that enables the actual implementation.

Is There a Modelling Framework for the Web?

- A modelling framework that is more suitable to the Web called **the Web Application Architecture Framework (WAAF)**, classifies concerns related to the development of Web Apps along two dimensions:
 - Row
 - Column
- The WAAF table is like a checklist that helps you organize all these different aspects of planning a web application.

The Web Application Architecture Framework (WAAF)

- **Each row** presents the perspective of the different participants in the WebE process:
 - business owners,
 - WebApp users,
 - information architects,
 - application architects,
 - developers,
 - and testers.

The Web Application Architecture Framework (WAAF)

- Each column represents a fundamental question about the web app, four categories that capture the main modelling domains:
 - **Structure (what):** data
 - **Behaviour (how):** Functions
 - **Location (where):** Network
 - **Pattern:** solutions to commonly encountered problems
- Each cell (where a row and column meet) **contains a specific plan, model, or description** related to that combination of person/view and aspect.
- The WAAF framework provides an indication of **which models are appropriate for which purpose and audience.**

The Web Application Architecture Framework (WAAF)

■ Examples

- **Business Owner + Structure:** A business entity relationship model (ERD).
- **User + Behavior:** A user interface flow model (a diagram showing how users will navigate through the web app).
- **Developer + Location:** A network deployment model (a diagram showing how the web app will be deployed on servers).

TABLE 6.1 WEB APPLICATION ARCHITECTURE FRAMEWORK (WAAF) MATRIX

	Structure (What)	Behavior (How)	Location (Where)	Pattern
Planning Architecture (Planner's Perspective)	List of things important to the business	List of processes the business performs	List of locations in which the business operates	Possible business models and patterns
Business Architecture (Business Owner's Perspective)	e.g., business entity relationship model	e.g., business process model	e.g., business entity location model	e.g., business model patterns
User Interface Architecture (User's Perspective)	e.g., user interface structure model	e.g., user interface flow model	e.g., user site map model	e.g., interface templates, navigation patterns
Information Architecture (Information Architect's Perspective)	e.g., information dictionary	e.g., information flow model	e.g., information node location model	e.g., information scheme patterns
System Architecture (System Architect's Perspective)	e.g., system functioning module/sub-module/server page structure	e.g., workflow model of module/submodule/server page	e.g., site mapping model of modules/submodules/server pages	e.g., design patterns, presentation styles
Web Object Architecture (Developers' Perspective)	e.g., physical object relationship	e.g., algorithms in source code	e.g., network deployment model	e.g., cots, components, code library
Test Architecture (Tester's Perspective)	e.g., test configuration	e.g., test procedure	e.g., test deployment	e.g., templates, standards of test document

How Does Modelling Relate to the WebE Process?

- Analysis and design actions are composed of the following WebE tasks:

Analysis Modeling (Chapter 7)

- A-1.** Decide whether a requirements model is needed.
- A-2.** Represent WebApp content.
- A-3.** Identify content relationships.
- A-4.** Refine and extend user scenarios.
- A-5.** Review user scenarios.
- A-6.** Create an interaction model for complex scenarios.
- A-7.** Refine interface requirements.
- A-8.** Identify functions.
- A-9.** Define constraints and performance requirements.
- A-10.** Identify database requirements.

How Does Modelling Relate to the WebE Process?

Design Modeling (Chapters 8 to 13)

- D-1.** Design the aesthetic for the WebApp.
- D-2.** Design the interface.
- D-3.** Design the navigation scheme.
- D-4.** Design the WebApp architecture.
- D-5.** Design the content and the structure that supports it.
- D-6.** Design functional components.
- D-7.** Design appropriate security and privacy mechanisms.
- D-8.** Review the design.

WAAF Modeling - Example

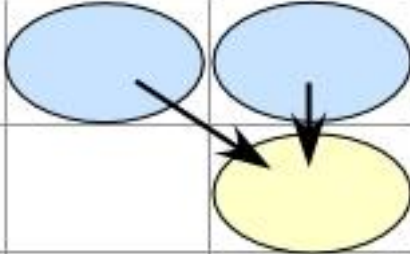
- Consider task **A-6, *Create an interaction model for complex scenarios.***
- As shown in the Figure next slide, the execution of this task involves:
 - understanding the relationship between user tasks and the content that is required for each task
 - It might also require information on business processes and information structures.

Analysis Modeling (Chapter 7)

- A-1. Decide whether a requirements model is needed.
- A-2. Represent WebApp content.
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- A-7. Refine interface requirements.
- A-8. Identify functions.
- A-9. Define constraints and performance requirements.
- A-10. Identify database requirements.

WAAF Modeling - Example

- Input to task A-6 typically includes broad business level information (i.e., the business owner's perspective) on both structure and behavior.
 - **structure** is defined by examining business entities and the relationships between them
 - **Behavior** is defined by examining business workflows.
- The outcome is a model that depicts how the users interact with the application (i.e., the application behavior) from the user's point of view.

	Structure (What)	Behaviour (How)	Location (Where)	Pattern
Planning Architecture (Planner's Perspective)				
Business Architecture (Business Owner's Perspective)				
User Interface Architecture (User's Perspective)				
Information Architecture (Information Architect's Perspective)				
System Architecture (System Architect's Perspective)				
Web Object Architecture (Developers' Perspective)				
Test Architecture (Tester's Perspective)				

WAAF Modeling - Example

- As a simple example of task A-6, consider increment 5, ***Information and ordering of monitoring services***, for **SafeHomeAssured.com**:
 - **Business-level structure (What)**: addresses the **specific information** that must be acquired to **set up a new user** for monitoring services.
 - the structure of the database that is used as a foundation for **customers** who use *SafeHome* monitoring services.
 - **Business-level behavior (How)**: addresses the **workflow that has been established to have a user register** for monitoring services.

WAAF Modeling - Example

- Task A-6 is about bridging the gap between the business's needs and the user's experience.
- It involves understanding the business's structure and processes and then translating that into a user-friendly interaction model for complex website tasks.

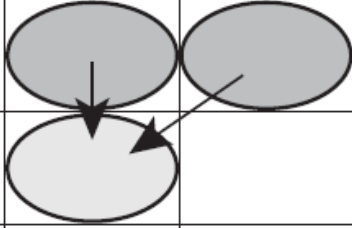
	Structure (What)	Behaviour (How)	Location (Where)	Pattern
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Business Architecture (Business Owner's Perspective)				
User Interface Architecture (User's Perspective)				
Information Architecture (Information Architect's Perspective)				
System Architecture (System Architect's Perspective)				
Web Object Architecture (Developers' Perspective)				
Test Architecture (Tester's Perspective)				

WAAF Modeling: Another Example

- The following figure shows the information required and the model created by design task **D-3**, ***Design the navigation scheme.*** In this case, the task involves
 - Input from the user's perspective—on both **structure** (i.e., the content and relationships between the content that the user interacts with) and **behavior** (i.e., the user's interaction model).
 - The output is a model of the application information architecture.

Design Modeling (Chapters 8 to 13)

- D-1.** Design the aesthetic for the WebApp.
- D-2.** Design the interface.
- D-3.** Design the navigation scheme.
- D-4.** Design the WebApp architecture.
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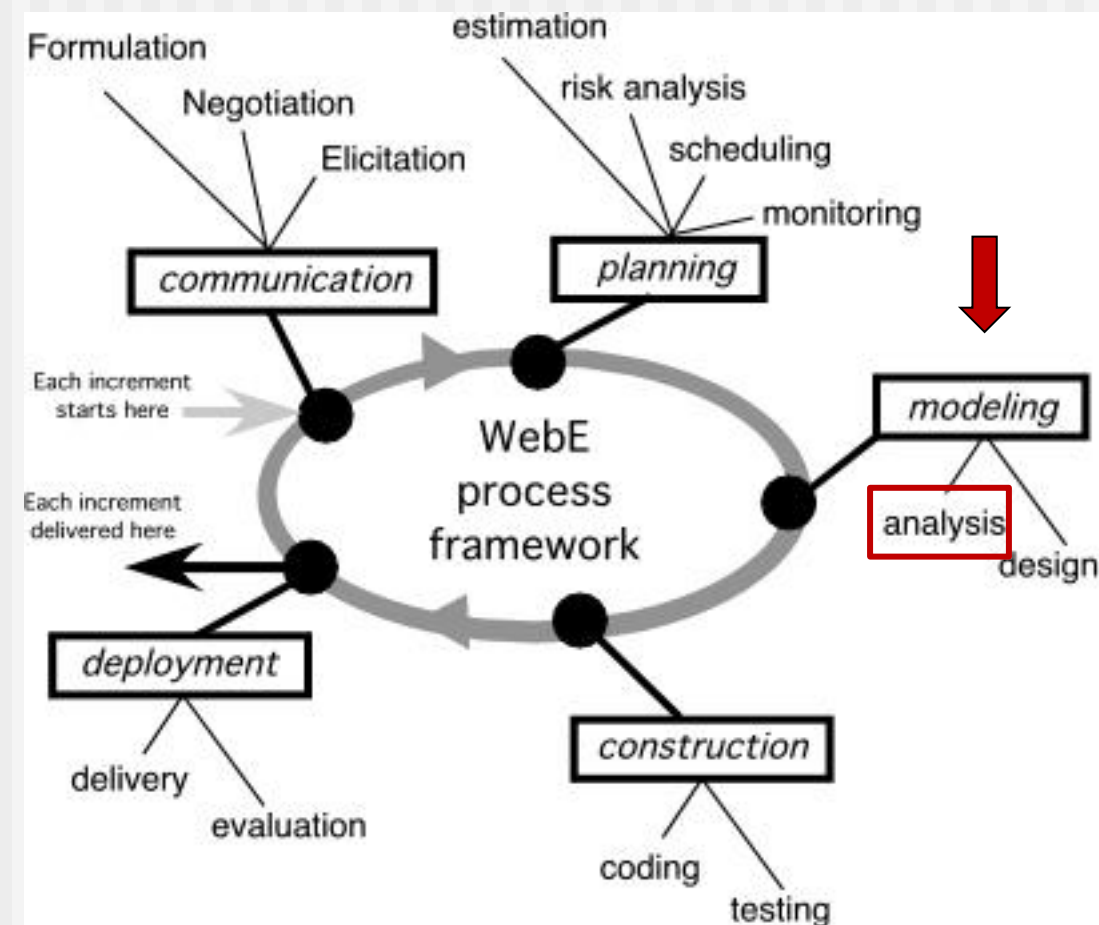
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User Interface Architecture (User's Perspective)				
Information Architecture (Information Architect's Perspective)				
System Architecture (System Architect's Perspective)				
Web Object Architecture (Developers' Perspective)				
Test Architecture (Tester's Perspective)				

WAAF Modeling - Another Example

- As a simple example of task D-3, consider increment 4, ***Space layout and security system design***, for **SafeHomeAssured.com**.
 - **User-level structure** addresses the specific information that must be specified by the user to create a space layout and how one data item relates to others.
 - **User-level behavior** specifies the flow of user actions that are required to create the space layout, coupled with system responses that flesh out the layout and recommend security monitoring devices.
- The output is translating the user's perspective into a clear and organized information architecture.



WebE Process Activities & Actions



Analysis Modeling

- Analysis modeling helps you to understand the detailed requirements that will allow you to **satisfy user needs**
- Analysis models look at **content, interaction, function, behavior, and the WebApp configuration**
- To determine the **how much analysis modeling** to do, examine the:
 - Size and complexity of the WebApp increment
 - Number of stakeholders (analysis can help to identify conflicting requirements coming from different sources)
 - Size of the WebE team
 - Degree to which members of the WebE team have worked together before (analysis can help develop a common understanding of the project)
 - Degree to which the organization's success is directly dependent on the success of the WebApp

Analysis Modeling

- But how do we actually *analyze* requirements? A useful way to answer this question is to consider the answers to four other key questions:
 1. What are the **inputs** (Information) to analysis modeling?
 2. What are the **outputs** from analysis?
 3. What analysis **tasks** can and should we carry out?
 4. What **tools** can we use to help us model?

What Are the **Inputs** to Analysis Modelling?

- The WebE process incorporates a communication activity that identifies **the information** that becomes input to analysis modelling.
 - This input includes stakeholders and user categories, the business context, defined informational and applicative goals, **general WebApp requirements, and usage scenarios.**
 - This information is represented in the form of natural language descriptions, rough outlines, sketches, and other informal representations.

Analysis Outputs

1. **Content (Information) model.** What information will be on the web app (e.g., product descriptions, images, blog posts). Content includes text, graphics and images, and video and audio data.
2. **Interaction model.** Describes the manner in which users interact with the WebApp (e.g., clicking buttons, filling out forms).
3. **Functional model.** Defines the operations that will be applied to WebApp content and describes other processing functions that are independent of content but necessary to the end user (e.g., calculate discounts, send email notifications).
4. **Configuration model.** Describes the environment and infrastructure in which the WebApp resides (e.g., what servers it will run on, what databases it will use).

Analysis Outputs

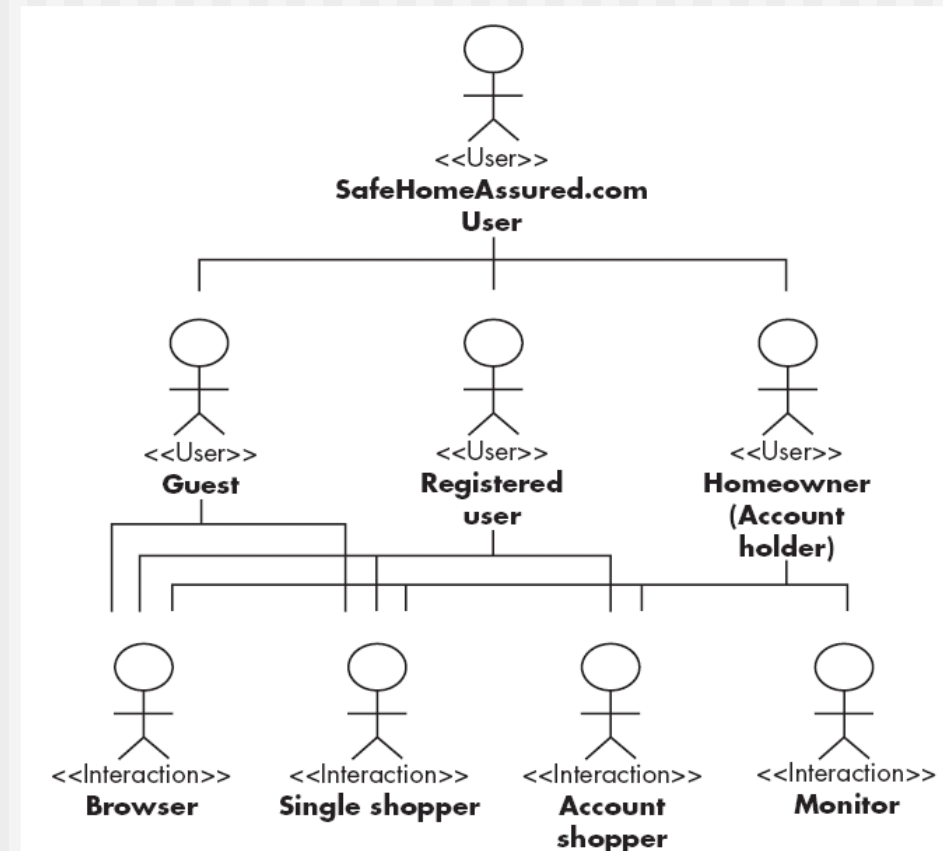
- Each of these models can be developed using a representation scheme (often called a “language”) that allows its intent and structure to be communicated and evaluated easily among members of the WebE team and other stakeholders.
- As a consequence, a list of key issues (e.g., errors, omissions, inconsistencies, suggestions for enhancement or modification, and points of clarification) are identified and acted upon.

Modeling Languages

- A *modeling language* (ML) incorporates a set of notations, terms, and/or symbols, as well as the rules for establishing associations between them
- A *modeling language* often has a formally structured representation as well as a set of graphical elements
- Some MLs are general purpose (e.g., UML) and others are more specific (e.g., WebML)

Understanding Users

- Crucial to understand your users!
- For each user class:
 - What is the user's overall objective?
 - What is the user's background?
 - How will the user arrive at the WebApp?
 - What characteristics does the user like and dislike?



The Content Model

- The content model contains structural elements that provide an important view of content requirements for a WebApp.
- These structural elements encompass content objects (e.g., text, graphical images, photographs, video images, audio) that are presented as part of the WebApp.

The Content Model

- Examples of content objects:
 - **External entities** (e.g., other systems, databases, people) that produce or consume information to be used by the WebApp
 - **Things** (e.g., reports, displays, video images) that are part of the information domain for the problem
 - **Occurrences or events** (e.g., a quote or an order) that occur within the context of a user's interaction with a WebApp
 - **Roles** (e.g., retail purchasers, customer support, salesperson) played by people who interact with the WebApp
 - **Organizational units** (e.g., division, group, team) that are relevant to an application
 - **Structures** (e.g., sensors, monitoring devices) that define a class of objects or related classes of objects

The Content Model

- The content model example of E-commerce website :
 - **Content types:** Products, categories, customer reviews
 - **Attributes:** Product name, description, price, images, category associations
 - **Relationships:** A product can belong to multiple categories; a customer review is linked to a specific product

The Content Model

- **Entity-Relationship Diagram (ERD)** is one of diagrams that can represent content models.
- Entity-Relationship Diagram (ERD): ERDs use rectangles for entities (content types) and diamonds for relationships. Lines connect entities, and cardinalities (e.g., one-to-many) are specified on the lines.

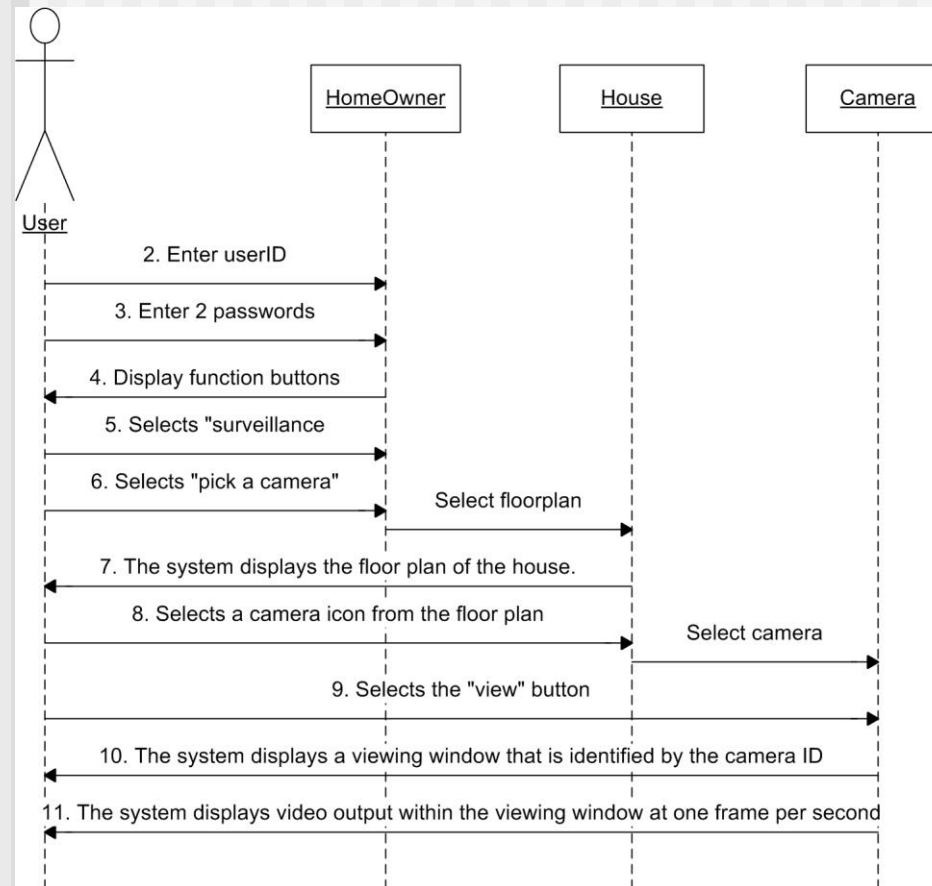
The Interaction Model

- The vast majority of WebApps enable a “**conversation**” between an end user and application functionality, content, and behavior. This conversation can be described using an interaction model.
- Can be represented using:
 - Use cases
 - Sequence diagrams
 - State diagrams
 - User interface prototypes

The Interaction Model

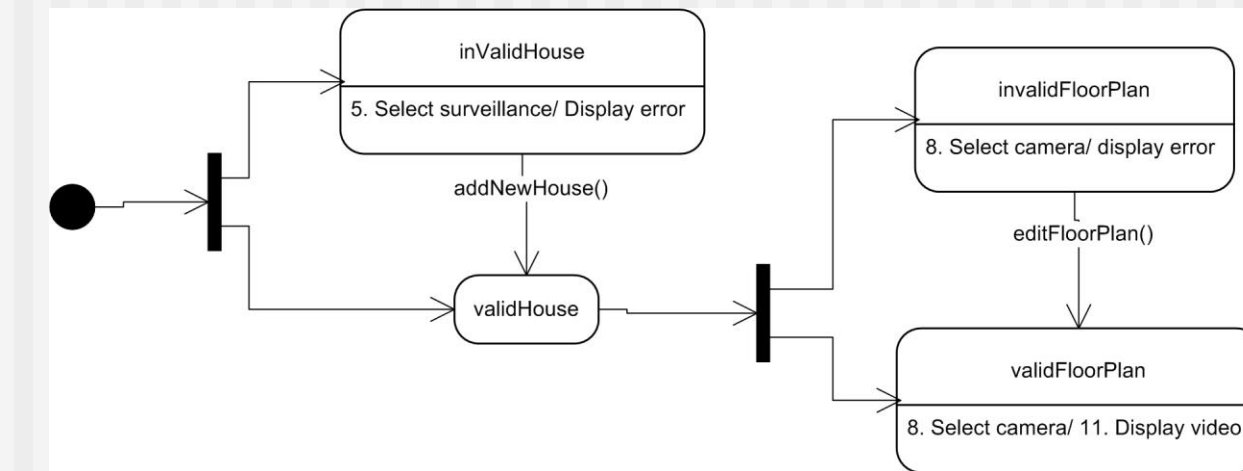
- In many instances, a set of use cases is sufficient to describe the interaction at an analysis level.
- However, when the sequence of interaction is complex and involves multiple analysis classes or many tasks, it is sometimes worthwhile to depict it using a more rigorous diagrammatic form.

The Interaction Model: Sequence Diagram



UML *sequence diagrams* describe how user actions collaborate with analysis classes (the structural elements of a system).

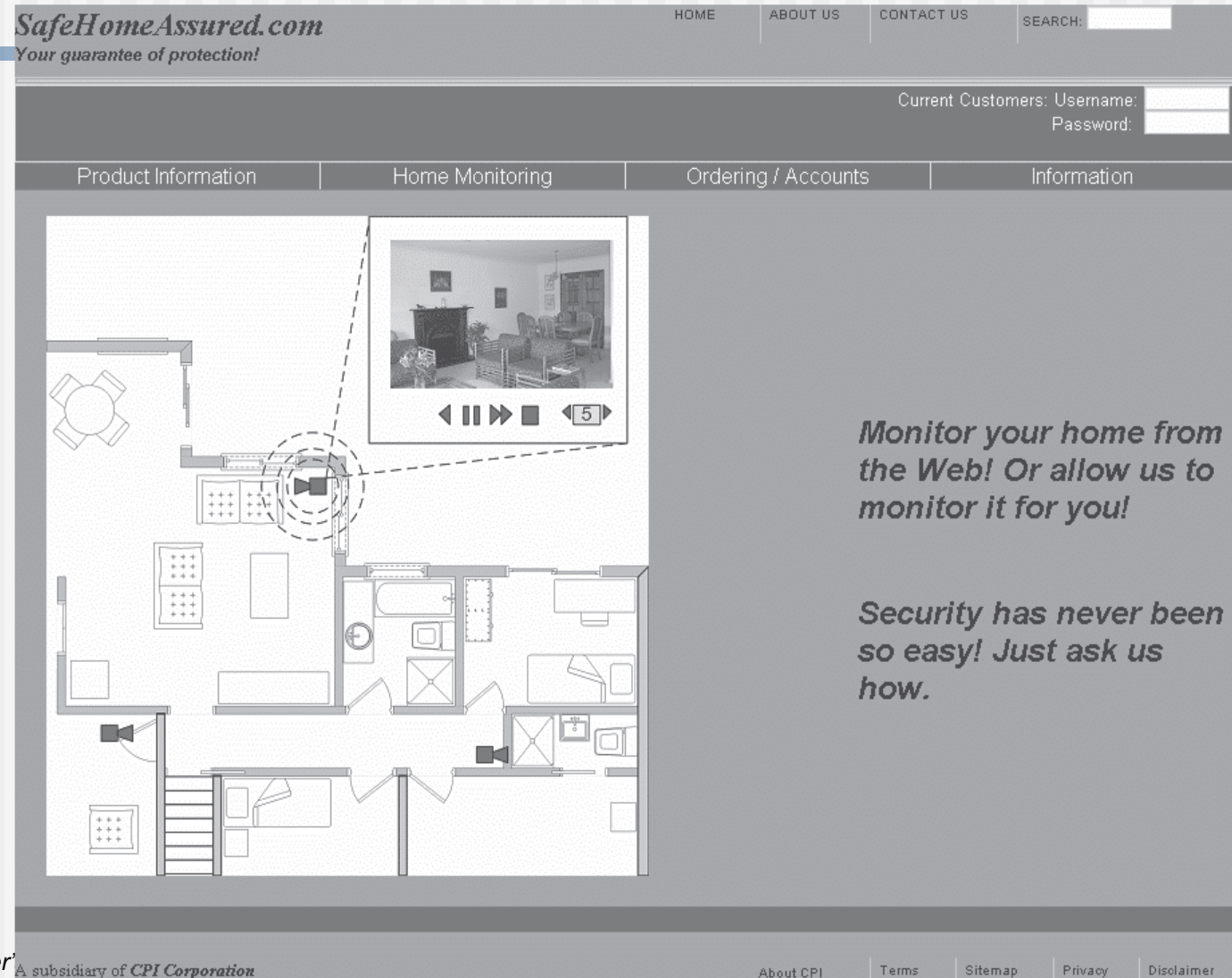
The Interaction Model: State Diagram



- UML *state diagrams* describe dynamic behavior of the WebApp as an interaction occurs.
- State diagrams are most useful when a user interaction triggers a change in the state of the WebApp—and hence changes the way in which it might react to a user.

The Interaction Model: User Interface Prototype

- It is a good idea to perform prototype during the creation of the analysis model.
- The sooner that a physical representation of a user interface can be reviewed, the higher the likelihood that end users will get what they want.
- The prototype may not activate these links, but they are provided to give the reviewer a feel for the manner in which navigation will be initiated.

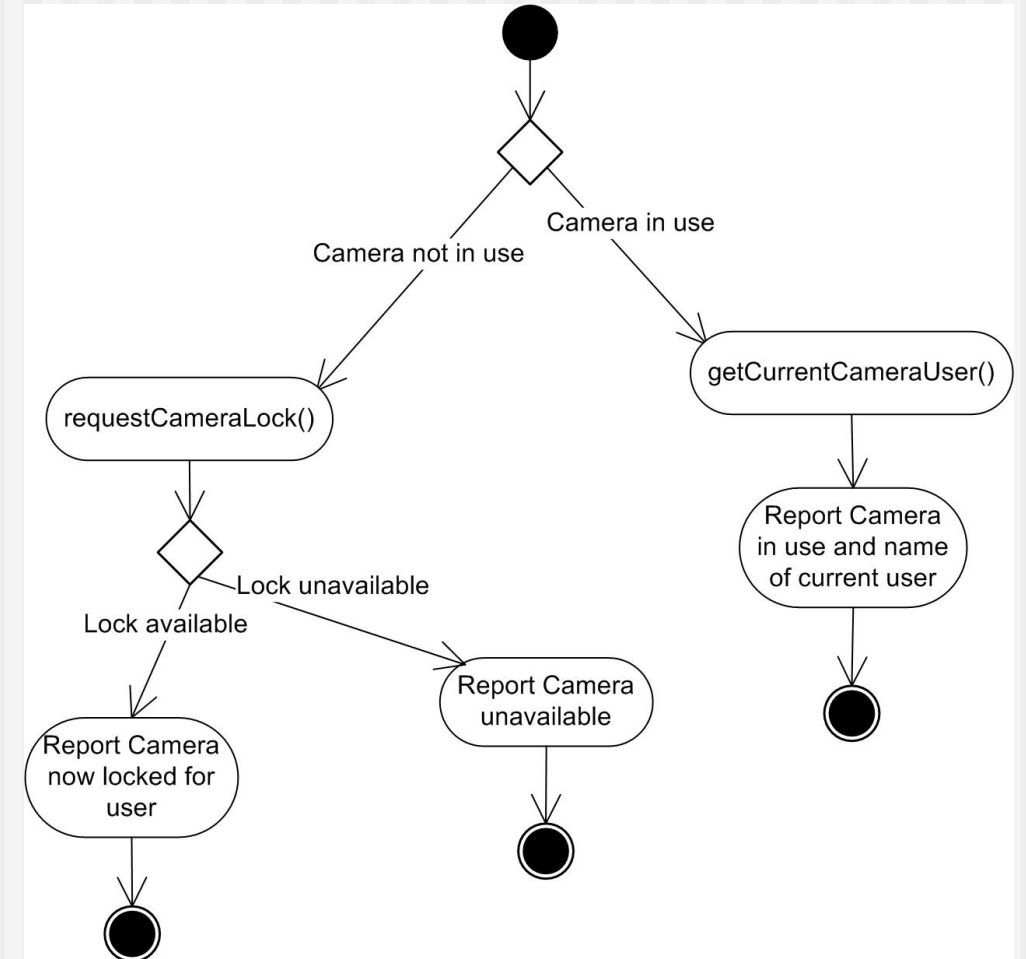


The Functional Model

- Addresses two processing elements of the WebApp, each representing a different level of procedural abstraction:
 - **user-observable functionality:** What the user sees and interacts with directly.
 - **Operations within analysis classes:** The internal workings of the web app that the user doesn't see
- The UML *activity diagram* can be used to represent processing details

The Functional Model: Activity Diagram

- Illustrates the processing flow and logical decisions within the flow.
 - The construction details indicate how these operations are invoked



The Configuration Model

- Configuration Model is considering the **environment** and **infrastructure** for a web application.
- There are many technical details to consider when planning a web app's configuration
 - Server hardware and operating system environments
 - Interoperability considerations on the server side (e.g., large database access, other IT applications, specialized communication protocols)
 - On the client side:
 - Local OS
 - Browser software
 - Client hardware variations

The Configuration Model

- In some cases, the configuration model is nothing more than a list of server side and client-side attributes.
- However, for more **complex WebApps**, a variety of configuration complexities (e.g., **distributing load among multiple servers, caching architectures**, remote databases, multiple servers serving various objects on the same Web page) may have an impact on analysis and design.

Relation-Navigation Analysis (RNA)

- As a web app grows larger and more complex, it becomes harder to manage **all the possible connections** and ensure users can easily find and **navigate what they need**.
- **Relation navigation analysis** is a specific type of analysis that focuses on **the relationships between entities** in a system and how data can be navigated between them.
- RNA is like creating a map for a web app, showing the connections between different elements and helping users navigate effectively.

Relation-Navigation Analysis

- The RNA approach is organized into five steps:
 - **Stakeholder analysis.** Identifies the various **user categories**, and establishes an appropriate stakeholder hierarchy
 - **Element analysis.** Identifies the **content objects and functional elements** that are of interest to end users
 - **Relationship analysis.** Describes the **relationships** that exist among the **WebApp elements**
 - **Navigation analysis.** Examines how **users** might access or **navigate** individual elements or groups of elements
 - **Evaluation analysis.** Assess the **cost and benefits** of implementing the identified relationships.

Relation-Navigation Analysis (Example)

- Building an e-commerce website for selling clothes. RNA might work:

1. Stakeholder Analysis:

- **Identify user classes:** Guest, Registered Customer, Admin.

2. Element Analysis:

- **Content objects:** Products, Categories, Shopping Cart, Orders, Customer Account.
- **Functional elements:** Search, Filter, Add to Cart, Checkout, Payment

Relation-Navigation Analysis (Example)

3. Relationship Analysis:

- **Product:**
 - Belongs to one or more Categories.
 - Has many Reviews.
 -
- **Category:**
 - Contains many Products.
 -
- **Order:**
 - Belongs to a Customer.
 - Contains many Products.
 -

Relation-Navigation Analysis (Example)

4. Navigation Analysis:

- Users should be able to:
 - Browse Products by Category.
 - Filter Products by attributes (size, color, price).
 - View Product details and Reviews.
 -

5. Evaluation Analysis:

- "Related Products" recommendations.
- "Customers who bought this also bought..." suggestions.
-

Relation-Navigation Analysis

■ Possible Diagrams for RNA:

- **Entity-Relationship Diagram (ERD)**
- **Site Map:** Shows the overall structure of the website and how pages are linked.
- **User Flow Diagram:** Shows the steps users take to complete specific tasks.
- **Wireframes:** Show the layout of individual pages and how navigation elements are placed.

Thank You